Unit 14 - Week 12

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Assignment 12

The due date for submitting this assignment has passed.
As per our records you have not submitted this assignment.

Due on 2019-10-23, 23:59 IST.

1) Which one of the following is not a factor contributing to the increased difficulty in software reliability measurement, as compared to hardware reliability measurement? *1 point*

○ a) The reliability improvement due to fixing a single bug depends on where the bug is located in the code.
○ b) The perceived reliability of a software product is observer-dependent.
○ c) The reliability of a software product keeps changing as errors are detected and fixed.
○ d) Software is invisible in contrast to hardware, though the effect of execution of software can be observed.

No, the answer is incorrect.
Score: 0

Accepted Answers:
d) Software is invisible in contrast to hardware, though the effect of execution of software can be observed.

2) Which one of the following can be said about the overall reliability behavior of a typical software during its life time? *1 point*

○ a) Initially increases, but later decreases
○ b) Initially decreases, but later increases
○ c) Increasing trend throughout the life cycle
○ d) Decreasing trend throughout the life cycle

No, the answer is incorrect.
Score: 0

Accepted Answers:
a) Initially increases, but later decreases

3) After a software is maintained, which one of the following types of testing involves repeating execution of tests that were already performed to ensure that the code modifications carried out during maintenance have not introduced additional faults? *1 point*

○ a) Beta testing
○ b) Functional testing
○ c) Bottom-up testing
○ d) Regression testing

No, the answer is incorrect.
Score: 0

Accepted Answers:
d) Regression testing

4) Which one of the following statements is not an objective of software verification? *1 point*

○ a) Ensuring that the software development steps are carried out correctly.
○ b) Ensuring that the correct software has been developed.
○ c) Achieving phase containment of errors.
○ d) Ensuring that the outputs produced at a development stage conform to the outputs of the previous phase.

No, the answer is incorrect.
Score: 0

Accepted Answers:
b) Ensuring that the correct software has been developed.

5) Which of the following is a practical use of reliability growth modelling? *1 point*

○ a) Determine the operational life of an application software
○ b) Determine when to stop testing
○ c) Incorporate reliability growth information while designing
○ d) Incorporate reliability growth information in the code

No, the answer is incorrect.
Score: 0

Accepted Answers:
b) Determine when to stop testing

6) Which one of the following is the objective of statistical testing? *1 point*

○ a) Use statistical techniques to design test cases
○ b) Apply statistical techniques to the results of testing to determine the number of latent bugs in the software
○ c) Estimate software reliability
○ d) Apply statistical techniques to the results of testing to determine how long testing needs to be carried out

No, the answer is incorrect.
Score: 0

Accepted Answers:
c) Estimate software reliability

7) What is the availability of a software with the following reliability figures? Mean Time Between Failure (MTBF) = 45 days, Mean Time To Repair (MTTR) = 5 hours. *1 point*

○ a) 10%
○ b) 20%
○ c) 90%
○ d) 99%

No, the answer is incorrect.
Score: 0

Accepted Answers:
d) 99%

8) On 4000 service invocations, a software failed on 10 occasions. What is its POFOD (Probability Of Failure On Demand)? *1 point*

○ a) 4%
○ b) 1%
○ c) 0.25%
○ d) 0.1%

No, the answer is incorrect.
Score: 0

Accepted Answers:
c) 0.25%

9) On a bug fix, the reliability of a software may actually decrease, rather than increase. Which one of the following explains this phenomenon? *1 point*

○ a) The fixed bug is in the core part
○ b) The fixed bug is in the non-core part
○ c) Bug-fix induces several other bugs
○ d) Bug-fix is not proper

No, the answer is incorrect.
Score: 0

Accepted Answers:
c) Bug-fix induces several other bugs

10) Which one of the following statements is FALSE? *1 point*

○ a) Every failure is caused due to a fault
○ b) A fault is a synonym for a bug
○ c) An error may or may not cause a fault
○ d) Every mistake committed by a programmer shows up as a failure

No, the answer is incorrect.
Score: 0

Accepted Answers:
d) Every mistake committed by a programmer shows up as a failure